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The Price of "Success of Children": How Does Intergenerational Education-Expectation Gap Affect Fertility Intention

A capstone project report submitted in partial fulfillment for the degree of Master of Social Sciences in the field of Population and Policy Analysis

By

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Declaration

We declare that this capstone project report represents our group's own work, except where due acknowledgement is made, and that it has not been previously included in a thesis, dissertation or report submitted to this University or to any other institutions for a degree, diploma or other qualification.

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Abstract

Against the backdrop of China's persistently low fertility rate and intensified educational competition, this study examines how the intergenerational gap in educational expectations affects families' willingness to have more child. Based on the data from the 2022 China Family Longitudinal Survey (CFPS), this study defines the intergenerational education-expectation gap as the difference between parents' expected educational level for their children and their own educational attainment. This study assesses the relationship between the gap and the disparity in fertility intentions through benchmark regression, heterogeneity analysis, mechanism testing, robustness testing, and instrumental variable analysis.

This study finds that the greater the intergenerational education-expectation gap, the smaller the family fertility intention gap, indicating that higher expectations for children's educational attainments may suppress parents' willingness to continue having children. This negative relationship is more pronounced among parents with lower educational attainment, middle-income families and the employed population. Meanwhile, only a relatively high level of educational expectation gap shows a significant inhibitory effect on fertility, indicating that there may be a threshold effect between the two. Mechanism analysis indicates that higher educational expectations may compress the space for families to continue having children by squeezing family economic resources and increasing the time and energy burden on parents. Instrumental variable analysis further supported the core results, but due to the limitations of cross-sectional data, this paper still maintains a cautious interpretation of causal relationships.

This study also indicates that low fertility is not only related to direct child-rearing costs, but is also influenced by educational competition, upward mobility anxiety, and intense parenting pressure. Therefore, in addition to providing economic and childcare support, policies that encourage childbirth should also pay attention to the structural pressure caused by educational competition and the uncertainty of families about future opportunities.

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1. Introduction

1.1 Background: Low Fertility and Educational Involution in China

The problem of low fertility rate in China is no longer merely a result of the family planning policy. According to the Statistical Bulletin, the total number of births in China in 2025 will be 7.92 million, with a birth rate of 5.63 per thousand and a natural population growth rate of -2.41% per thousand (National Bureau of Statistics of China, 2026). Although the multi-child policy has been in place for many years, existing research shows that the relaxation of the policy has not automatically translated into a stronger willingness to have a second or third child (Jing et al., 2022; Ren et al., 2026). Therefore, this means that China's low fertility rate cannot be simply understood as the residual effect of early policy restrictions. It also needs to explain why the existing parents continue to view having an additional child as a decision involving long-term parenting investment and uncertain returns, thus contributing to the result of low fertility intentions.

Education is one of the most important pressures behind this decision. Chinese families' investment in children's education is not merely ordinary consumption; it is closely linked to children's future opportunities, the family's social position, and expectations of intergenerational mobility. Existing studies show that household education expenditure, private tutoring, and shadow education have clear competitive and stratified characteristics in China (Wei, 2024; Zhang & Bray, 2021). In this environment, parents are not simply calculating how much it costs to raise one more child. They are also assessing whether they can provide the educational resources needed to prevent downward mobility and, ideally, support upward mobility. Educational involution therefore turns childrearing into a source of subjective mobility anxiety: the more children are regarded as carriers of the family's future class position, the more decisions about additional childbearing are constrained by educational expectations.

Existing studies on low fertility have largely emphasized objective constraints, including education expenditure, housing costs, opportunity costs, and child-rearing costs. Research on educational competition has also begun to show how private tutoring and educational investment reshape family resource allocation (Wei, 2024; Meng et al., 2026). However, these explanations pay less attention to a more relative dimension, which is the gap between parents' own educational position and the educational achievements they expect their children to attain. For existing parents, additional fertility intention may be restrained not only by the price of education itself, but also by the subjective standard of what educational attainment a child should reach. The effect of educational involution on additional fertility intention therefore comes not only from external educational costs, but also from the pressure generated by parents' expectations for intergenerational educational mobility.

1.2 Concept Definition: Intergenerational Education-Expectation Gap and Additional Fertility Intention

To capture this subjective dimension, this study introduces the concept of the intergenerational education-expectation gap. It refers to the gap between the educational attainment of parents themselves and the expected educational attainment for their children to achieve. Unlike the educational attainment of parents, the gap in intergenerational educational expectations reflects a family's current socioeconomic status, while the gap in intergenerational educational expectations reflects relative expectations for the future. It shows how far parents believe their children should move beyond the educational position of the parental generation. Against the backdrop of intensified educational competition, this expectation is not merely a general family aspiration. It is linked to children's future competitiveness, the maintenance of family social status, and the pressure to avoid downward mobility.

This conceptual shift clarifies that the focus of this study is not educational attainment itself, but the relative education-expectation gap. Parents with the same educational attainment may hold very different expectations for their children's educational achievement. Conversely, parents with different educational attainment may experience similar pressure if they expect their children to achieve substantial educational upward mobility. The key issue, therefore, is not whether parents are highly educated, but whether they understand child-rearing as a long-term task that requires children to move beyond the parental generation. Compared with absolute educational attainment, the intergenerational education-expectation gap better captures the subjective mobility anxiety behind educational competition.

The outcome of interest in this study is additional fertility intention among existing parents. Because the effective sample consists of parents who already have at least one child, fertility intention in this study refers to parents' expected or desired future childbearing beyond their current children, rather than realized fertility behavior or first-birth intention. This distinction is important because additional fertility intention captures the stage at which families evaluate whether they are willing and able to take on another long-term child-rearing responsibility. In the Chinese context, where child-rearing is closely tied to educational competition, this outcome provides a useful lens for examining how perceived educational pressure enters reproductive decision-making among already-parent families.

1.3 Research Question and Hypotheses

Based on this problem construction, the research question of this study is: How does the intergenerational education-expectation gap affect additional fertility intention among Chinese parents who already have at least one child? This question links China's low fertility problem to the subjective expectation pressure embedded in educational involution. This study does not simply revisit whether educational costs suppress fertility. Instead, it asks whether parents' expectations for children's educational mobility become a relative source of pressure on the desire for additional children. The intergenerational education-expectation gap thus provides a concrete entry point for understanding the relationship between educational competition, class anxiety, and low additional fertility intention among existing parents.

Centering on this study question, this study proposes three core hypotheses. H1: A larger intergenerational education-expectation gap is associated with lower additional fertility intention among existing parents. H2: This negative relationship is expected to differ across families with different resource constraints, especially across parental educational attainment and income positions. H3: The relationship between the intergenerational education-expectation gap and additional fertility intention may be related to household education investment pressure and parents' time and energy constraints. These hypotheses point to a common argument: educational involution affects additional fertility intention not only through rising educational expenditure, but also through the subjective pressure parents attach to children's educational mobility.

1.4 Contribution

This study makes four contributions. First, at the conceptual level, it shifts the analysis of education and additional fertility from absolute educational attainment to the relative education-expectation gap. Existing studies often understand educational factors through parental education, children's education expenditure, or family educational resource input. In contrast, the focus of this study is on the gap between parents' own educational attainment and their expectations for their children's education. This shift makes it possible to capture the sources of stress that are difficult to observe through traditional educational variables. That is to say, parents may not only assess their children's burden based on their own educational level or current educational costs, but also on whether their children are expected to achieve educational mobility after their parents' generation.

Not only that, at the theoretical level, this study advances the discussion of external education prices to subjective mobility anxiety. Educational expenses are an important constraint on fertility decisions. However, in a highly competitive educational environment, educational pressure is also reflected in parents' subjective judgment of their children's future class status, the risk of failure in educational competition, and the possibility of intergenerational mobility, among other aspects. By placing educational involution, social mobility pressure, and fertility suppression within the same analytical framework, this study argues that low additional fertility intention among existing parents is not only a response to cost constraints, but may also reflect the entry of class mobility anxiety into family decision-making.

Third, at the empirical level, this study uses CFPS data to examine whether the intergenerational education-expectation gap is associated with additional fertility intention among existing parents in China. This allows the study to move beyond broad claims about education pressure and to test whether relative expectation pressure is systematically related to fertility plans for additional children. The empirical analysis further considers whether this relationship differs across family conditions, which helps identify the groups for whom education-related fertility pressure is more pronounced.

Fourth, at the methodological level, this study operationalizes subjective educational pressure by constructing the intergenerational education-expectation gap as the core explanatory variable. This variable is measured through the difference between parents' own educational attainment and their expectations for children's educational attainment. Unlike research designs that rely only on education expenditure or parental education, this measure incorporates the relative expectation of the educational position children are expected to reach into testable empirical analysis. This study further combines heterogeneity analysis and mechanism testing to examine where the association is stronger and whether education investment pressure and time-energy constraints constitute possible pathways. Through this design, the study not only tests whether education-expectation gaps are related to additional fertility intention among existing parents, but also advances empirical understanding of how educational competition is transformed into fertility pressure on additional childbearing.

2. Literature Review

2.1 Education, Social Mobility, and Family Investment

Education is widely regarded as the main channel for intergenerational status acquisition and social class reproduction. Breen and Goldthorpe's (1997) theory of formal rational behavior suggests that family education decisions are fundamentally driven by "relative risk aversion," meaning that the primary goal of the family is not necessarily to achieve upward mobility, but to ensure that their children receive sufficient education credentials to avoid downward mobility and at least maintain the family's current socio-economic status (Breen and Goldthorpe, 1997). Goldthorpe (2014) combined empirical sociological research with economic indicators to demonstrate the contradictory role of education in social stratification. He emphasized that for privileged or middle-class families, education investment is primarily used as a defense mechanism to protect their descendants from intergenerational displacement, rather than just as an opportunity equalizer. In a fiercely competitive society, this fear of class decline is a powerful catalyst for parental investment. In China, education is deeply rooted as the ultimate path to success, and the scarcity of elite educational resources has triggered profound "class anxiety" among parents. Driven by a strong fear of social class backwardness, families continuously increase their investment and expectations in education. Even at the cost of sacrificing other family resources, it is important to ensure children's competitive advantage and construct a multidimensional barrier to prevent downward mobility (Yan et al., 2026). Therefore, the extreme emphasis on children's education is not only a pursuit of academic excellence, but also a defensive strategy against social displacement (Breen&Goldthorpe, 1997; Goldthorpe, 2014).

2.2 Quantity-Quality Trade-Off and Fertility Decisions

Becker and Lewis' (1973) "Quantity Quality Trade-off" theory provides a fundamental microeconomic framework for understanding the mechanisms behind family fertility decisions and human capital investment. They believe that in situations where family resources such as time, energy, and income are severely limited, parents face an inherent trade-off between the "quantity" (number of children) and "quality" (level of resource investment per child) of their children.

Specifically, an increase in the number of children naturally dilutes available resources, thereby increasing the shadow price or marginal cost of raising a high-quality child. Lundholm and Ohlsson (2002) extended this analysis by introducing parental time constraints. They believe that an increase in income may increase the demand for both the quantity and quality of children. However, if the demand for quality grows faster, the rising cost per child may still suppress quantity. Therefore, when parents have high educational expectations for their children, they will adopt a "quantity for quality" strategy to ensure upward social mobility (Juhn et al., 2015). To avoid diluting limited family wealth and parental energy, families often choose to voluntarily limit the size of their children, concentrating scarce resources on one child to maximize the child's growth outcomes and future competitiveness (Li et al., 2008).

2.3 From Parental Education to Educational Expectations

Early demographic studies mainly focused on the direct inhibitory effect of parents' absolute education level on family fertility intentions (Blossfeld & Huinink, 1991). Numerous empirical studies have shown that an increase in years of education typically leads to a significant decrease in total fertility rates and a delay in the time of first childbirth (Baizan & Nie, 2024; Brand & Davis, 2011; Gustafsson, 2001). From the perspective of opportunity cost in microeconomics, an increase in education level increases an individual's potential wage return in the labor market, leading to a significant increase in the time cost of sacrificing to raise children, which directly suppresses the fertility willingness of highly educated groups. In the process of population transition in China, with the comprehensive expansion of higher education, the highly educated population has reduced the overall number of children in order to balance career development and family investment (Lavely&Freedman, 1990).

However, relying solely on objective absolute education level is no longer sufficient to fully explain the complex micro fertility decisions of contemporary Chinese families. In recent years, the academic community has gradually realized that the focus of research perspectives must shift towards subjective educational expectations. The latest micro empirical research shows that after controlling for relevant variables, the direct negative impact of parents' own education level on their willingness to have a second child is not significant. The improvement of education mainly increases parents' micro education expectations for their children, indirectly inhibiting family fertility decisions (Zhong, 2026). This confirms that subjective educational expectations have a stronger correlation than objective absolute education in explaining modern low fertility behavior. Many scholars have also drawn similar conclusions in their analysis of Chinese household tracking survey data, namely that educational expectations play a central bridging role between individual socioeconomic status and fertility decisions (Wang, 2023). In the intense social competition and changes, highly educated groups often have a stronger motivation to maintain social status, which translates into highly institutionalized academic expectations for the next generation, bringing enormous pressure on family resource allocation at the micro level (Bian & Xiong, 2023).

Although the above research confirms the inhibitory effect of educational expectations on fertility decisions, most empirical literature still has limitations and only focuses on exploring the absolute educational expectations of parents (Liu, 2015). In the current highly competitive social environment, providing children with higher education has become a common cultural consensus, and simply measuring absolute expectations is no longer effective in distinguishing the real pressures that different families are under (Zhong & Xie, 2024). Jiang et al. (2022) pointed out that the status anxiety and parenting burden felt by parents are not fundamentally due to high expectations themselves, but rather to the sharp gap between these high expectations and the actual social status of the family.

Based on the limitations of previous research, this article officially introduces intergenerational education expectation gap as the core explanatory variable, defining it as the relative gap between parents' expected education level for their children and their actual education level. Compared to traditional absolute measurement methods, measuring this difference can more sensitively and accurately quantify parents' class anxiety in the intergenerational transmission process (Zhong & Xie, 2024). The larger the difference, the deeper the social class gap that the family needs to cross, reflecting parents' extreme desire for upward mobility of their offspring and their profound fear of class sliding (Breen & Goldthorpe, 1997). In order to bridge the huge gap between expectations and reality, families are often forced to adopt aggressive defensive education investment strategies, which directly trigger strong wealth crowding-out and energy occupation effects (Yan et al., 2026; Bian&Xiong, 2023). Therefore, when faced with dual constraints of budget and time, families completely lose the space to expand their fertility scale and can only adopt defensive strategies of limiting fertility.

2.4 Wealth Crowding-Out Effect

Wei (2024) shows that rising household education expenditure constitutes an important financial constraint for Chinese families. Driven by intense social competition and an increasingly stratified labor market, parents are forced to invest heavily in supplementary education and high-quality educational resources to ensure intergenerational upward mobility for their offspring. Research has shown that the proportion of family education expenditure in the total income of Chinese households is too high, resulting in serious budget constraints (Doepke & Zilibotti, 2019). From a microeconomic perspective, when parents attempt to bridge the gap between their actual socioeconomic status and their high educational expectations for their children, the marginal economic cost of raising each additional child grows exponentially. The rigid expansion of education expenditure has directly triggered a profound wealth crowding-out effect. Therefore, the financial resources that families could have used to conceive and raise a second child are greatly diluted and occupied by investments in the education of the first child (Wei, 2024). This wealth crowding-out effect exhibits significant heterogeneity among different socio-economic classes. Lei and Shen (2015) also pointed out that low and middle-income families suffer the most severe wealth crowding-out, as their rigid education expenditures account for a much higher proportion of the total household budget compared to affluent families. In order to alleviate this heavy economic burden, the Chinese government has introduced the "double reduction" policy, aimed at regulating after-school tutoring, reducing the cost of compulsory education, and attempting to unleash the suppressed fertility potential of ordinary families (Meng et al., 2026). However, sociological analysis suggests that as long as the fundamental intergenerational gap in educational expectations remains unresolved, parents will continue to seek alternative channels for educational investment (Doepke & Zilibotti, 2019). In addition, this wealth crowding-out effect is not evenly distributed throughout society, leading to a phenomenon increasingly conceptualized as middle-class crowding in sociological literature (Wang, 2023). Although high-income families have sufficient capital buffers to absorb rising education premiums without sacrificing overall living standards, while low-income families may largely withdraw from elite education competition due to absolute resource barriers, the middle class finds themselves in structural difficulties (Lei & Shen, 2015). Due to a strong desire to maintain social status and prevent downward mobility of offspring, middle-income parents often internalize their highest relative educational expectations. However, their family budget has strict limits. Therefore, they are forced to disproportionately allocate limited financial resources to education investment, leaving minimal fiscal flexibility for additional births (Doepke & Zilibotti, 2019). Therefore, the squeeze of the middle class highlights how the wealth squeeze effect becomes most severe when high liquidity desires conflict with strict budget constraints.

2.5 Energy Occupation Effect

In addition to strict economic restrictions, high educational expectations impose severe demands on parents' time and energy, resulting in a huge energy consumption effect and severely suppressing their desire to have children. Contemporary sociology of education emphasizes a structural shift towards strengthening parenting practices, which requires parents to invest unprecedented time, emotional labor, and physical effort in their children's daily academic supervision and extracurricular development (Zhang et al., 2022). In order to translate high educational expectations into actual academic success, parents must be deeply and continuously involved in their children's learning process. This high demand parenting model greatly increases the overall burden of parenting and directly exacerbates the common time shortage among people of childbearing age (Zhang et al., 2022). The energy occupation effect is particularly detrimental to dual income families. Young working parents find themselves caught in a serious structural conflict between demanding professional responsibilities and highly intensive parenting demands (Jiang et al., 2022). A longitudinal study of dual income families in China suggests that attempting to meet these elevated parenting standards can lead to long-term role overload and severe parental burnout. This state of physical and mental exhaustion is a key mediating mechanism that directly and negatively affects the willingness to have a second child (Jiang et al., 2022). In addition, mothers bear a disproportionate share in this energy occupation as they traditionally have the primary responsibility for overseeing family education. The intense work family conflict experienced by working mothers is one of the most decisive factors leading to a sustained decline in maternal fertility intentions. Ultimately, the time squeeze and energy occupation caused by the pursuit of high educational expectations leave families with insufficient personal resources to manage the need for additional children, thereby reinforcing the defense strategy against birth restrictions.

2.6 Educational Involution and Resource Pressure

From the perspective of the mechanism of wealth crowding-out, educational internalization has triggered severe peer and ratchet effects in family education investment. Under intense macro competition pressure, families are forced to engage in a comprehensive resource consumption war in order to make up for their intergenerational education expectations. Empirical studies in economics have shown that high levels of educational competition not only lead to rigid increases in explicit financial costs such as off campus tutoring and high-quality school district properties, but also transform this high standard of educational consumption into a universal social norm (Lei & Shen, 2015). When the surrounding families in the social network are constantly increasing their investment in education, some families must adopt a follow-up strategy to ensure that their children are not marginalized, even when facing overall budget constraints. This passive additional defensive education investment leads to an exponential increase in the overall marginal financial cost of raising a single child (Doepke & Zilibotti, 2019). Therefore, in an environment of internal competition, extremely high educational expectations no longer remain at the psychological level, but directly transform into extremely heavy and almost incompressible rigid family debts. This systematic financial overdraft completely weakens the objective economic foundation for most ordinary families to expand their fertility, making them unable to bear the direct cost of raising another child.

From the perspective of the mechanism of energy consumption, educational involution has profoundly reshaped the parenting culture of contemporary society, pushing the traditional parenting model to an extreme and highly intensive parenting stage (Zhang et al., 2022). Under the current education selection mechanism, providing only economic support from the family is considered far from enough, and parents are recognized by the social system as the core responsible persons and daily managers of their children's education. In order to meet extremely high educational expectations and cope with the pressure of internal competition, parents not only need to invest a large amount of funds, but also have to forcibly transfer a large amount of private time originally used for self recovery, career development, or leisure and entertainment to deep accompanying study, high-intensity academic supervision, and complex academic path planning (Zhong & Xie, 2024). This competitive pressure completely publicizes and internalizes the private time of families, leading to deep parenting fatigue and fear of childbirth among childbearing age groups when faced with the decision to have a second child due to severe depletion of physical and mental resources. Therefore, educational involution, as a powerful structural force, has led to a critical threshold of family energy consumption caused by poor expectations, greatly suppressing the psychological willingness of childbearing age groups to expand their fertility scale.

Currently, educational involution has become a macro structural background that cannot be ignored in understanding the micro fertility decisions and educational investment behaviors of Chinese families. Sociological research on education points out that educational involution is essentially an objective environment in which high-quality educational resources are relatively scarce and social mobility channels are gradually tightening. Families engage in fierce competition to compete for relative competitive advantages (Zhang, 2025b). In this highly competitive macro social context, the expectations of families for the academic achievements of their offspring are constantly being pushed higher. Multiple micro empirical studies have shown that this external macro competitive pressure directly and exponentially amplifies the aforementioned wealth crowding-out and energy occupation effects, making micro family fertility decisions face more stringent external constraints (Zhong & Xie, 2024).

2.7 Socioeconomic Heterogeneity and Resource Buffering

Despite the enormous pressure exerted by the macro level educational internalization environment on all families, families of different socioeconomic status exhibit significant class heterogeneity in specific fertility decisions when faced with intergenerational education expectations of equal intensity. Population economics and social stratification theory profoundly point out that the inherent resource endowment of a family is the most core moderating variable that regulates the relationship between educational expectations and fertility intentions (Breen & Goldthorpe, 1997). The socio-economic status of a family not only determines the scale of financial resources it can mobilize, but also fundamentally determines its micro ability to resist macro competition risks and absorb education cost premiums.

For low and middle income or low educated families with limited resources, they often exhibit extremely fragile economic and energy situations when dealing with high educational expectations. According to the relative risk aversion assumption in social mobility theory, when humans make investment decisions in family education, their primary rational motivation is often not to achieve class advancement, but to try their best to avoid their offspring experiencing class decline (Breen & Goldthorpe, 1997). Families with limited resources themselves lack sufficient idle capital pools and rich social capital networks, and their total budget has strong hard constraint characteristics. At the same time, parents of such families often work in fixed and inflexible occupations, facing more rigid time poverty. When these disadvantaged families have extremely high expectations for their children to cross or at least maintain their current social class, the most effective and even the only rational strategy they can mobilize is to reduce the number of children (Yan et al., 2026). Faced with the extremely high shadow prices caused by educational competition, low socioeconomic status families are unable to cope with wealth crowding-out through external financial borrowing or increasing total income. Therefore, by compressing extremely limited family resources to the extreme and concentrating them highly and exclusively on a single offspring, it has become their optimal defense against further decline in social status (Lei & Shen, 2015). Empirical research based on micro panel data such as the China Family Panel Studies (CFPS) has confirmed that low socioeconomic status families, when facing the same educational expectation pressure, experience a significant and substantial inhibitory effect on their fertility intentions compared to the advantaged class group (Lei & Shen, 2015; Zhong, 2026). For them, having a second child means a severe dilution of existing scarce family resources, which directly threatens the class competitiveness of their existing children. Therefore, they are more likely to adopt extreme defensive strategies that limit the number of children they have.

On the contrary, high-income and highly educated families with high socioeconomic status not only have more abundant financial buffering capacity when facing the same intergenerational education expectations gap, but also can effectively resolve the energy consumption crisis through market-oriented means (Doepke & Zilibotti, 2019). Firstly, in terms of wealth crowding-out, the overall budget size of high-income families is sufficiently large, and although the high premium of high-quality educational resources also constitutes household expenditure, it is far from reaching the bottom line of household financial security. They can spend these high education costs without significantly reducing the overall quality of family life and savings for having a second child. In addition, high-income families have strong purchasing power over time and can transfer their energy burden by purchasing high-quality market-oriented parenting services and educational alternative resources (Doepke & Zilibotti, 2019). For example, they can hire professional household workers to undertake heavy daily care work, hire senior private tutors to take over high-intensity academic tutoring tasks, and even rely on professional educational consulting agencies to complete complex college planning. The ability to deeply market and outsource intensive parenting processes has enabled high-level parents to successfully free up a significant amount of time from tedious parenting tasks, greatly alleviating work family conflicts and time poverty (Doepke & Zilibotti, 2019). This significant resource buffering capacity and market-oriented time transfer ability enable upper class families to maintain high educational expectations and cope with intense educational competition while still retaining high reproductive decision-making flexibility. They do not need to make extreme trade-offs between the quality and quantity of their children, so that they still have a stronger practical feasibility and willingness to have a second child in the context of a sustained decline in the macro fertility rate.

2.8 Research Gap and Study Contribution

Based on previous research, the academic community has fully established the important consensus that macro level educational competition and micro level high educational expectations have a significant negative impact on family fertility decisions (Doepke&Zilibotti, 2019; Zhong, 2026). However, there are still academic gaps in the existing literature when exploring this issue. In terms of conceptual measurement dimensions, most empirical studies mainly focus on static measurements of objective education costs or absolute parental education expectations, lacking systematic quantification and in-depth exploration of the relative gap indicator of intergenerational education expectations, making it difficult to accurately capture the real class anxiety and relative deprivation of contemporary families. In terms of empirical methods, some existing studies have failed to effectively address endogeneity issues caused by omitted variables or bidirectional causality when evaluating the relationship between educational characteristics and fertility intentions, and have not thoroughly examined the specific micro transmission mechanisms behind the suppression of fertility (Zhong, 2026). In order to compensate for the limitations of the above research, this article will use nationally representative micro panel data from the CFPS (2022) to formally establish intergenerational education expectation gap as the core explanatory variable. In terms of empirical strategy, this article will conduct benchmark analysis using ordinary least squares method and introduce instrumental variable estimation models to effectively overcome potential endogeneity bias, thereby systematically quantifying the true inhibitory effect of this expectation gap on fertility intention. At the same time, it will deeply examine the specific operational paths of the two core mechanisms of wealth crowding-out and energy occupation.

3. Development of Hypotheses

3.1 Main effect of intergenerational education-expectation gap and fertility intention (H1)

According to Breen and Goldthorpe's (1997) theory of relative risk aversion, the primary motivation for families in educational investment decisions is not to achieve class advancement, but to prevent their offspring from experiencing class decline. When parents' educational expectations for their children are significantly higher than their actual level of education, this gap in expectations constitutes a strong signal of class anxiety, in which families must cross higher educational thresholds than the previous generation in order to maintain or even improve their current social status (Goldthorpe, 2014). However, under the premise of limited family resources (time, money, energy), the quantity quality trade-off theory proposed by Becker and Lewis (1973) states that parents must make a trade-off between the "quantity" and "quality" of their children. Bridging the gap in intergenerational educational expectations requires extremely high resource investment, which directly increases the marginal cost of raising each child (Liu&Li, 2015). Therefore, when parents perceive a significant gap in expectations, in order to ensure that their existing children can gain sufficient competitiveness to prevent downward mobility, families tend to voluntarily limit their fertility and highly concentrate limited resources on fewer children (Yan et al., 2026). Based on this, this study proposes:

H1: A larger intergenerational education-expectation gap is associated with lower additional fertility intention among existing parents.

3.2 Moderating Effect of Parental Educational Attainment

Although the gap in intergenerational educational expectations may have a fertility suppression effect in all families, the strength of this effect varies depending on the parents' own educational background. The social stratification theory and cultural capital framework indicate that families with different levels of education will experience different types of stress when faced with the same expectation gap (Bourdieu, 1986).

For parents with lower levels of education, setting high educational goals for their children means crossing a huge educational distance. Unlike parents with higher education, they are naturally endowed with rich cultural capital, familiarity with academic systems, and mature educational networks, while parents with lower education often lack these intangible resources (Lareau, 2024). When they expect their children to achieve significant upward mobility, they face steeper information barriers and must make greater compensatory efforts to navigate complex educational pathways. Psychological anxiety, information loss, and perceived difficulties in ensuring academic success are greatly magnified (Lai et al., 2022).

Therefore, in order to minimize this huge intergenerational gap and reduce the high risk of educational failure, parents with lower levels of education are more likely to adopt highly defensive strategies. If there is no natural advantage of cultural capital, their most feasible choice is to concentrate their limited cognitive, emotional, and parental bandwidth on one child(Lei & Shen, 2015). On the contrary, parents with higher education have greater confidence and ability to guide their children's academic development, making the gap in expectations feel more manageable. Based on this, this study proposes:

H2**:** The negative effect of the gap is stronger among parents with lower education levels.

3.3 The mediating mechanism of wealth crowding-out and energy occupation (H3)

The intergenerational education expectation gap itself does not directly reduce the willingness to have children, but plays a role through two specific micro transmission paths: wealth crowding-out effect and energy occupation effect. The first path is wealth crowding-out effect, which stems from the high economic costs required to bridge the gap in expectations. Parents are forced to invest heavily in supplementary education programs such as extracurricular tutoring, high-quality school district housing, and specialized training in order to ensure that their children have at least the same or higher level of education as themselves (Wu, 2026). These education expenditures account for a disproportionately high proportion of the total household income, creating serious budget constraints and systematically squeezing financial resources that could have been used for childbirth and raising a second child (Doepke&Zilibotti, 2019). When education expenditure shows rigid growth, families lose the ability to expand their fertility scale on an objective economic level. The second path - energy consumption effect - focuses on the consumption of time and emotional resources. The contemporary high-intensity intensive parenting model requires parents (especially mothers) to invest unprecedented time and emotional labor in academic supervision, home school communication, and academic planning (Xu et al., 2023). In order to translate high educational expectations into actual academic achievements, parents are forced to compress their time for self recovery, career development, and leisure and entertainment, falling into a long-term state of role overload and time poverty (Jiang et al., 2022). The severe depletion of physical and mental resources directly weakens the psychological willingness and practical feasibility of reproductive age groups to have children again. Based on this, this study proposes:

H3**:** The gap suppresses fertility through two main mechanisms: a) increased economic investment; b) increased time and energy investment.

4. Data and Methodology

4.1 Data Source

This study uses data from the China Family Panel Studies 2022 (CFPS 2022), a nationally representative longitudinal survey project conducted by the Institute of Social Science Survey at Peking University. CFPS provides rich information on individuals, households, children, education, employment, income, family structure, and fertility-related attitudes, making it suitable for examining the relationship between educational expectations and fertility intention in contemporary China.

The empirical dataset combines information from the CFPS Person and Childproxy files. The Person file supplies respondents' demographic and socioeconomic characteristics, including age, gender, hukou status, health, employment, income, and fertility expectations. The Childproxy file records the educational attainment that parents expect for their children. Linking the two files allows us to compare child-focused expectations with the respondent's own education.

The unit of analysis is the individual respondent. Although child educational expectation is originally recorded at the child level, the final regression dataset is constructed at the individual level. Therefore, child-level educational expectation information is first aggregated to the household level and then merged back into the person-level dataset using the household identifier.

4.2 Data Processing and Sample Construction

Preparing the analytical sample required four main steps.

First, special missing-value codes in CFPS were cleaned. CFPS uses negative values such as "not applicable," "don't know," "refuse to answer," and other non-substantive responses. These codes were recoded as missing values before any calculation or regression analysis. This step is necessary because otherwise these special codes would be treated as real numeric values and would distort the construction of variables such as fertility gap and education-expectation gap.

The Person file serves as the base file. It contains the respondent's age, gender, hukou status, health status, job satisfaction, income, employment status, fertility expectation, and current family structure. The expected number of children is taken from this file, while the number of children currently recorded in the household is used to measure realized family size.

We processed the Childproxy file before merging it with the Person file. A household may report information for several children, so a direct merge would create repeated person records. We therefore summarized child-level educational expectations at the household level. Both the mean and the maximum expectation within a household were considered; the household-level measure used in the main analysis was then matched to the respondent to calculate the intergenerational education-expectation gap.

Because the Childproxy file only contains households with children, the effective sample is limited to already-parent families. Therefore, this study examines additional fertility intention rather than first-birth intention or fertility intention among all Chinese adults.

After this aggregation, the household summary was linked to the Person file by household ID. Each regression uses complete cases for the variables included in that specification. Consequently, the number of observations changes when additional controls are introduced.

4.3 Variable Construction

4.3.1 Dependent Variable: Fertility Intention Gap

For each respondent, we calculate the difference between the number of children they expect to have and the number currently observed. We refer to this difference as the fertility intention gap:

$$Fertility\ Intention\ Gap_i = Expected\ Children_i - Actual\ Children_i$$

Expected $Children_i$ is the respondent's reported expected number of children, whereas Actual $Children_i$ is the current number of children recorded for that respondent or household.

This measure retains more information than a yes-or-no question about another child. Its magnitude indicates how far the respondent's current number of children falls short of their stated expectation.

As a separate check, we also estimate a model for second-child status. This outcome allows us to compare the continuous fertility intention gap with a concrete second-birth measure.

4.3.2 Core Independent Variable: Intergenerational Education-Expectation Gap

We define the intergenerational education-expectation gap as the difference between the educational attainment parents expect for their children and the attainment parents have achieved themselves. A larger value therefore represents a longer expected educational advance across generations.

Formally, it is defined as:

$$EEG_i = Child's\ Expected\ Education_h - Parent's\ Education_i$$

Here, EEG_i denotes the raw intergenerational education-expectation gap for respondent i. Child's Expected $Education_h$ is the household-level expectation for children's attainment, and Parent's $Education_i$ is the respondent's own attainment. Because the original educational-attainment and education-expectation variables are categorical, the categories are converted into approximate completed years of schooling using the coding scheme below. This conversion makes it possible to calculate a continuous intergenerational education-expectation gap.

Coding Scheme for Educational Attainment

Educational attainment category	Assigned years
Illiterate	0
Primary school	6
Junior high school	9
Senior high school	12
College/Vocational	15
Bachelor's degree	16
Master's degree	19
Doctoral degree	22

Note: The conversion expresses categorical educational attainment as approximate completed years of schooling.

The unstandardized education-expectation gap is therefore measured in estimated years of schooling. A positive value indicates that the educational attainment expected for the child exceeds the parent's own attainment by that number of years, while a negative value indicates a lower expected attainment.

For the regression models, the raw gap is converted into a standardized score:

$$ZEEG_i = \frac{EEG_i - \overline{EEG}}{SD(EEG)}$$

We standardize EEG to make effect sizes comparable across specifications. The coefficient on ZEEG can be read as the change associated with a one-standard-deviation increase in the raw education-expectation gap. Centering and scaling are also useful in the interaction and instrumental-variable models, where variables otherwise operate on different numerical ranges. Because the schooling-year assignments are approximate, standardization should be understood as a change of scale; it does not remove measurement error or alter statistical significance by itself.

4.3.3 Control Variables

The baseline specifications adjust for the following observed characteristics:

X_i = {age, gender, hukou status, health status, job satisfaction, income, employment status}

Age accounts for the respondent's position in the life course, while gender captures possible differences in fertility preferences and family responsibilities. Hukou status reflects institutional differences in access to urban services and educational resources. Self-rated health and job satisfaction represent aspects of individual well-being. The extended specification adds household income and employment status, and province fixed effects capture broad regional differences in fertility culture, education systems, and socioeconomic conditions.

4.4 Empirical Strategy

4.4.1 Baseline OLS Model

We begin with an ordinary least squares model that relates the intergenerational education-expectation gap to the fertility intention gap:

$$Fertility\ Intention\ Gap_i = \beta_0 + \beta_1 ZEEG_i + \beta_2 X_i + \varepsilon_i$$

In this equation, Fertility Intention Gap_i is the dependent variable, $ZEEG_i$ is the standardized intergenerational education-expectation gap, X_i contains the observed controls, and ε_i is the error term.

The parameter of interest is β_1. A negative estimate would mean that respondents with larger education-expectation gaps also tend to report smaller gaps between expected and current family size, conditional on the included controls.

The reported analysis uses two specifications. Model 1 includes the core demographic and institutional controls. Model 2 adds income, employment status, and province fixed effects:

$$Fertility\ Intention\ Gap_i = \beta_0 + \beta_1 ZEEG_i + \beta_2 X_i + \gamma_p + \varepsilon_i$$

Here, γ_p represents province fixed effects.

4.5 Heterogeneity Analysis

We next assess whether the relationship varies with parents' educational attainment. The sample is first divided into low-, medium-, and high-education groups, and the baseline model is estimated within each group:

$$Fertility\ Intention\ Gap_i = \beta_0 + \beta_1 ZEEG_i + \beta_2 X_i + \varepsilon_i$$

These subgroup estimates show whether the negative association is concentrated among parents with less education, who may face greater difficulty mobilizing the resources needed to meet ambitious expectations for their children.

We then fit an interaction specification to test the differences jointly:

$$\begin{aligned} &Fertility\ Intention\ Gap_i \\ &= \beta_0 + \beta_1 ZEEG_i + \beta_2 MediumEdu_i + \beta_3 HighEdu_i \\ &+ \beta_4 (ZEEG_i \times MediumEdu_i) + \beta_5 (ZEEG_i \times HighEdu_i) \\ &+ \gamma X_i + \varepsilon_i \end{aligned}$$

The interaction terms provide a direct statistical test of whether the slope for the education-expectation gap differs from that of the low-education reference group.

4.6 Mechanism Analysis

The study further examines two mechanisms: wealth crowding-out and time/energy occupation.

4.6.1 Wealth Crowding-Out Mechanism

The first mechanism suggests that higher educational expectations increase household educational expenditure, which in turn reduces fertility intention. This mechanism is tested through a three-step mediation framework.

First, the total effect of the education-expectation gap on fertility intention is estimated:

$$Fertility\ Intention\ Gap_i = \beta_0 + \beta_1 ZEEG_i + \beta_2 X_i + \varepsilon_i$$

Second, the effect of the education-expectation gap on educational expenditure is estimated:

$$log\_eexp_i = \alpha_0 + \alpha_1 ZEEG_i + \alpha_2 X_i + \mu_i$$

Third, educational expenditure is added to the fertility model:

$$Fertility\ Intention\ Gap_i = \delta_0 + \delta_1 ZEEG_i + \delta_2 log\_eexp_i + \delta_3 X_i + \nu_i$$

If educational expenditure is negatively associated with fertility intention and the coefficient of the education-expectation gap decreases after adding educational expenditure, this would provide evidence for the wealth crowding-out mechanism.

Second, to provide a more rigorous validation of the wealth crowding-out mechanism based on Becker's "quantity-quality trade-off" theory, this study supplements the mediation analysis with a budget constraint test. Specifically, the full sample is partitioned into three strata—low-income, middle-income, and high-income groups—based on household income percentiles. The baseline model is then estimated separately for each income stratum:

$$\begin{aligned} &Fertility\ Intention\ Gap_i \\ &= \beta_0 + \beta_1 ZEEG_i + \gamma X_i + \varepsilon_i \ \ (estimated\ by\ income\ group) \end{aligned}$$

By comparing the magnitude and significance of the coefficient β_1 across these strata, the analysis examines the financial constraint. This heterogeneous cross-validation would provide robust evidence that educational expectations suppress fertility intention through actual resource pressure.

4.6.2 Time and Energy Occupation Mechanism

The second proposed pathway concerns parents' time and energy. Meeting high educational expectations may involve homework supervision, tutoring arrangements, school choice, and educational planning, all of which compete with time available for another child.

To explore this pathway, we estimate the baseline model separately for employed and non-employed respondents:

$$Fertility\ Intention\ Gap_i = \beta_0 + \beta_1 ZEEG_i + \beta_2 X_i + \varepsilon_i$$

A stronger negative coefficient among employed respondents would be consistent with work-family time pressure playing a role. Because employment is only an indirect indicator of available time and energy, this comparison is interpreted as exploratory.

4.7 Robustness Checks

We use three additional checks to assess how sensitive the findings are to measurement and model choice.

First, the continuous education-expectation gap is replaced with low-, medium-, and high-gap categories based on its sample distribution. This specification checks whether the relationship is gradual or concentrated among respondents with especially large gaps:

$$\begin{aligned} &Fertility\ Intention\ Gap_i \\ &= \beta_0 + \beta_1 MediumGap_i + \beta_2 HighGap_i + \beta_3 X_i + \varepsilon_i \end{aligned}$$

The low-gap category is omitted as the reference group. A negative coefficient only for the high-gap category would be consistent with a threshold pattern rather than a linear relationship.

Second, we estimate a logit model for second-child status:

$$logit[P(SecondChild_i = 1)] = \beta_0 + \beta_1 ZEEG_i + \beta_2 X_i$$

This specification asks whether the findings depend on using a continuous intention measure rather than a second-child outcome.

Third, we use the community-level average education-expectation gap as an instrument in a sensitivity analysis. The motivation is that respondents living in the same community may be exposed to a shared educational climate that shapes their expectations. The approach remains cautious: community expectations could also be related to local school quality, housing costs, or neighborhood socioeconomic conditions, so the exclusion restriction cannot be verified directly.

4.8 Summary of Methodological Design

The empirical analysis therefore proceeds from the baseline association to subgroup comparisons, alternative variable coding, mechanism-oriented tests, and an instrumental-variable sensitivity analysis.

Taken together, these steps show where the relationship is strongest and how it changes under different specifications, while keeping the interpretation appropriately cautious for cross-sectional data.

5. Results and Interpretation

5.1 Baseline Regression Results

Table 1: Baseline Regressions on Fertility Gap

Variable	Model 1	Model 2
Education-expectation gap	-0.063*** (0.019)	-0.064** (0.025)
Controls		
age	0.026*** (0.002)	0.036*** (0.003)
gender	0.016 (0.036)	-0.062 (0.048)
Health status (1-5 scale)	0.025* (0.013)	0.031* (0.018)
Job satisfaction (1-5 scale)	0.016** (0.005)	-0.032 (0.023)
Logged income (ln_inc)		0.007 (0.016)
Employed		0.067 (0.104)
Hukou controls	Yes	Yes
Province FE	No	Yes
Observations	5,859	3,268
R2	0.045	0.096
Adj. R2	0.043	0.085

*Notes: Standard errors in parentheses. Dependent variable is the fertility intention gap. Hukou controls are included but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Table 1 reports the baseline OLS estimate of the association between the intergenerational education-expectation gap and the fertility intention gap. In Model 1, the model controls for core demographic characteristics, including age, gender, self-assessed health status, and hukou status. The estimated coefficient for the intergenerational education-expectation gap is -0.063 and is statistically significant at the 1% level. Given that the expectation-gap variable is standardized, this coefficient implies that a one-standard-deviation increase in the intergenerational education-expectation gap is associated with a 0.063-unit decrease in the fertility intention gap. This means that when parents' educational expectations for their children are higher than their own educational attainment, the gap in their intention to have children is lower.

Under this model setting, the performance of control variables also provides rich empirical information. For example, the estimated coefficient of the age of the respondents is significantly positive at the statistical level of 1% ($\beta = 0.026$, $p<0.01$), which shows that with the increase of age, the gap between the ideal number of children and the actual number of children of an individual tends to widen.

Because age may capture both life-cycle effects and cohort differences in fertility norms, we conduct an age-stratified robustness check. The sample is divided into respondents under 40 and those aged 40 and above. The results show that the negative association between the education-expectation gap and fertility intention gap is stronger among respondents under 40 ($\beta = -0.115$, $p < 0.001$), while the coefficient is smaller and only weakly significant among respondents aged 40 and above ($\beta = -0.045$, $p < 0.10$). This suggests that the main association is concentrated among parents who are still in the core stage of additional childbearing decision-making.

Finally, it is worth noting that the R^2 value of the baseline models ranges from 0.04 to 0.09. As is typical in cross-sectional micro-demographic studies, this implies that individuals' reproductive intentions are highly complex and driven by many unobservable psychological factors. Nevertheless, the significant inhibitory effect of our core variables remains robust.

To reduce concerns about omitted macro-environmental variables or family financial characteristics, Model 2 employs a stricter empirical testing strategy. Building on model 1, the model further introduces the family-level logarithmic annual income and individuals' employment status, whether employed or not, as multi-dimensional economic covariates, and fully controls for provincial fixed effects. Due to missing questionnaire data when combining cross-module variables, the effective observation sample size for model 2 has been reduced to 3,268. However, thanks to the addition of multi-dimensional control variables, the explanatory power of the model has increased significantly, and the adjusted R-squared (Adj. R^2) increased from 0.043 to 0.085.

At the same time, the estimated coefficient for the intergenerational education expectation difference of the core explanatory variable remains highly stable. Its numerical value is slightly adjusted to -0.064 and remains significant at the 5% statistical level, and the standard error is 0.025. In summary, the result in Table 1 clearly reveals that a larger intergenerational education-expectation gap is associated with lower willingness to expand family size among existing parents.

5.2 Heterogeneity and Subgroup Analysis

In order to reveal the unequal impact of educational involution on different social classes, this section investigates the heterogeneous distribution of the expectation gap's suppressive effect by introducing subgroup regressions and interaction terms.

5.2.1 Heterogeneity by Educational Attainment

Table 2: Heterogeneity by Educational Attainment

Variable	Low education	Medium education	High education
Education-expectation gap	-0.187*** (0.060)	-0.095* (0.046)	-0.044 (0.038)
Controls			
age	0.027*** (0.003)	0.027*** (0.003)	0.021*** (0.003)
gender	-0.129 (0.080)	0.067 (0.060)	0.064 (0.054)
Health status (1-5 scale)	-0.030 (0.025)	0.030 (0.020)	0.068** (0.022)
Job satisfaction (1-5 scale)	0.007 (0.012)	0.012 (0.009)	0.021* (0.009)
Hukou controls	Yes	Yes	Yes
Observations	1,234	2,252	2,373
R2	0.058	0.050	0.038
Adj. R2	0.053	0.047	0.034

*Notes: Standard errors in parentheses. Dependent variable is the fertility intention gap. Hukou controls are included but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

First of all, this study examines the differences of families with different levels of educational attainment in intergenerational educational anxiety. The subgroup regression results in Table 2 show that the suppressive effect of the education-expectation gap on fertility intentions shows a decreasing trend with the increase in parents' education. Specifically, the low-education group experiences the strongest negative association, which is highly significant at the 1% level (the estimated coefficient is -0.187), followed by the medium-education group (-0.095, p<0.10), and for the higher education group, this negative effect is decreased in absolute magnitude and loses statistical significance. This step-wise decreasing pattern may suggest that, due to limited cultural capital and educational resources, when less-educated parents try to break down class barriers and have high educational expectations for their children, the relative effort and resource mobilization difficulty may increase. This resistance to reality can easily exhaust their limited household resources, thereby reducing their intention to have a second child.

To verify the robustness of these cross-class differences under a statistical framework, Table 3 presents the Interaction Model. Using the low-educated group as the reference category, the interaction coefficients for the expectation gap with medium-education and high-education exhibit positive signs (0.094 and 0.124, respectively). However, as these interaction terms fail to reach conventional statistical significance, Hypothesis 2 is not formally confirmed under this joint framework. The step-wise attenuation observed in the subgroup models (Table 2) should therefore be interpreted as suggestive rather than conclusive evidence. While the positive coefficients of the interaction terms descriptively align with the hypothesis that higher parental education mitigates the suppressive effect of the expectation gap, the lack of statistical significance implies that the cross-strata variance is not sufficiently pronounced to clear strict joint testing thresholds in this cross-sectional sample. Consequently, H2 is only partially supported and warrants further longitudinal investigation.

Table 3: Interaction Model: Education-Expectation Gap x Education Group

Variable	**Model**
Education-expectation gap	-0.191*** (0.057)
edu_group: medium	-0.201** (0.076)
edu_group: high	-0.229** (0.084)
Education-expectation gap x medium	0.094 (0.072)
Education-expectation gap x high	0.124 (0.070)
Controls	
age	0.025*** (0.002)
gender	0.021 (0.036)
Health status (1-5 scale)	0.024* (0.013)
Job satisfaction(1-5 scale)	0.015** (0.005)
Hukou controls	Yes
Observations	5,859
R2	0.046
Adj. R2	0.044

*Notes: Standard errors in parentheses. Dependent variable is the fertility intention gap. The reference category for edu_group is the low-education group. Hukou controls are included but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

5.2.2 Robustness Check: Impact Differences across Varying Expectation Gaps

Table 4: Robustness Check: Trichotomized Education-Expectation Gap

Variable	Model
Education-expectation gap: medium	-0.010 (0.043)
Education-expectation gap: high	-0.128*** (0.047)
Controls	
age	0.026*** (0.002)
gender	0.017 (0.036)
Health status (1-5 scale)	0.025** (0.013)
Job satisfaction (1-5 scale)	0.016*** (0.005)
Hukou controls	Yes
Observations	5,859
R2	0.045
Adj. R2	0.043

*Notes: Standard errors in parentheses. Dependent variable is the fertility intention gap. The reference category is the low education-expectation gap group. The education-expectation gap is recoded into three groups based on the 33rd and 67th percentiles. Hukou controls are included but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

The robustness test in Table 4 examines the suppressive effect of educational expectations on fertility, and whether there are differences under different expectations. This study reconstructs the standardized expectation gap variable into three categories of variables: low, medium, and high, according to the 33rd and 67th percentiles. The regression results show structural differences in the inhibition of fertility across different expectation gaps. Compared with the low expectation-difference reference group, the moderate expectation-difference group does not exhibit a significant fertility inhibition effect (the coefficient is only -0.010 and not statistically significant). Only the high expectation-gap group shows a statistically significant negative association (the coefficient is as high as -0.128, $p<0.01$). This result shows that there are structural differences in the impact of educational expectations on fertility decision-making. Moderate educational expectations reflect rational family planning. Its resource consumption is still within the controllable range and may not substantially hinder the intention to have children. However, when the expectation gap reaches a high level, the strong anxiety regarding upward social mobility may push the family to overuse time and financial resources, thus losing some flexibility to raise another child under the harsh constraints of reality.

5.3 Mechanism Analysis: A Resource Constraint Perspective

Exploring the possible pathways through which the education-expectation gap is associated with lower additional fertility intention is an important part of this study. Traditional Baron & Kenny path analysis is vulnerable to the endogeneity bias of baseline wealth when dealing with the absolute expenditure of cross-sectional data. Therefore, this section returns to Becker's theory of "quantity-quality trade-off" and conducts dual cross-validation through strict budget and time constraints.

5.3.1 Descriptive Evidence on Resource Constraints: The Wealth Crowding-Out Hypothesis

Table 5: Exploratory Test of Education Expenditure as a Mediator

Variable	Step 1: Total effect	Step 2: Mediator	Step 3: With mediator
Education-expectation gap	-0.063*** (0.019)	-0.085 (0.049)	-0.057** (0.019)
Controls			
log_eexp			-0.016** (0.005)
age	0.026*** (0.002)	-0.021*** (0.004)	0.025*** (0.002)
gender	0.016 (0.036)	-0.083 (0.090)	0.003 (0.036)
Health status (1-5 scale)	0.025 (0.013)	0.065* (0.032)	0.026* (0.013)
Job satisfaction(1-5 scale)	0.016** (0.005)	-0.022 (0.014)	0.017** (0.005)
Hukou controls	Yes	Yes	Yes
Observations	5,859	5,654	5,654
R2	0.045	0.017	0.046
Adj. R2	0.043	0.016	0.044

*Notes: Standard errors in parentheses. Step 2 uses log_eexp as the mediator. Hukou controls are included but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Table 6: Heterogeneity by Income Group

Variable	Low income	Middle income	High income
Education-expectation gap	-0.037 (0.062)	-0.117*** (0.035)	-0.024 (0.048)
Controls			
age	0.026*** (0.006)	0.041*** (0.004)	0.044*** (0.007)
gender	-0.150 (0.114)	-0.040 (0.067)	0.001 (0.100)
Health status(1-5 scale)	0.022 (0.037)	0.043 (0.026)	0.051 (0.040)

Variable	Low income	Middle income	High income
Job satisfaction (1-5 scale)	-0.071* (0.042)	-0.002 (0.033)	-0.012 (0.056)
Hukou controls	Yes	Yes	Yes
Observations	827	1,677	805
R2	0.035	0.067	0.058
Adj. R2	0.025	0.063	0.049

*Notes: Standard errors in parentheses. Dependent variable is the fertility intention gap. Middle income combines medium-low and medium-high quartiles. Hukou controls are included but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

To explore the wealth crowding-out mechanism, we initially tested a traditional Baron & Kenny three-step mediation model using the logarithm of absolute education expenditure (log_eexp) as a mediator (Table 5). However, Step 2 reveals that the impact of the education-expectation gap on absolute education expenditure is counterintuitively negative and not statistically significant ($\beta = -0.085$, $p > 0.10$). So the traditional causal mediation chain is not supported. This highlights the inherent limitations of cross-sectional data, as a family's baseline wealth pool strongly confounds the measurement of absolute education expenditure.

Given the broken formal mediation chain, we refrain from claiming strict causality. Instead, we discuss the budget constraints (Table 6) as a descriptive background. The results of income heterogeneity deviate from a simple linear resource-constraint expectation. The results do not show that the lowest-income group is subjected to the strongest fertility suppression, but reveal a significant middle-class squeeze phenomenon. The core coefficients of low-income and high-income groups are -0.037 and -0.024, respectively, and there is no statistical significance. In contrast, middle-income families face a severe and highly significant negative correlation ($\beta = -0.117$, $p < 0.01$).

This unexpected deviation provides crucial insight into the contemporary Chinese context. Unlike high-income families who can absorb educational costs without sacrificing quality of life or low-income families who may largely withdraw from elite education competition, the middle class is in a rigid resource dilemma. They harbor a strong desire for upward social mobility and attempt to consolidate their social status through educational investment, yet they are bound by strict family budget ceilings. Consequently, to meet the high educational expectations for their first child, middle-income families may feel pressured to allocate their limited financial resources to improve the quality of their child. This pattern is consistent with a wealth crowding-out interpretation: for middle-income families, maintaining high educational expectations may reduce the perceived financial room for additional childbearing.

5.3.2 Energy Occupation Mechanism

Table 7: Mechanism Analysis: Time/Energy Constraint by Employment Status

Variable	Employed	Not employed
Education-expectation gap	-0.073*** (0.021)	0.002 (0.045)
Controls		
age	0.026*** (0.002)	0.026*** (0.004)
gender	-0.011 (0.040)	0.089 (0.101)
Health status (1-5 scale)	0.026 (0.015)	0.023 (0.025)
Job satisfaction (1-5 scale)	-0.019 (0.016)	0.012 (0.007)
Hukou controls	Yes	Yes
Observations	4,869	990
R2	0.039	0.091
Adj. R2	0.037	0.085

*Notes: Standard errors in parentheses. Dependent variable is the fertility intention gap. Hukou controls are included but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

In addition to financial constraints, we further explore the time and energy occupation mechanism by using the respondents' employment status (employed vs. non-employed) as an objective proxy (Table 7).

The subgroup regression results reveal a clear structural contrast. In the "employed group" model, which contains 4,869 observations, the intergenerational education-expectation gap has a strong negative association with the family's fertility intention gap. Its estimated coefficient is -0.073, and it is highly significant at the statistical level of 1%. In contrast, in the non-employed group model, containing 990 observations, the core coefficient of the education-expectation gap drops to 0.002, which is not only numerically negligible but also statistically insignificant. This empirical comparison between groups suggests that the negative association of the education-expectation gap on the decision-making of having a second child does not affect all families in the same way, but is mainly concentrated among "employed groups" facing strict time constraints. This finding provides a microeconomic and sociological explanation for the "Energy Occupation Mechanism" .

In addition, this finding forms a coherent logical link with the wealth crowding-out channel discussed in Section 5.3.1. Under Becker's "quantity-quality trade-off" framework, the pursuit of children's "quality" requires not only substantial financial investment (Table 5 and Table 6), but also requires parents to invest a lot of time and energy (Table 7). The dual constraints of financial budget and time allocation may reduce the flexibility required for additional childbearing.

Since employment status is only an indirect proxy for time and energy constraints, this result should be interpreted as exploratory rather than definitive evidence of the mechanism.

5.4 Sensitivity Analysis using Instrumental Variables

Although the benchmark OLS model has controlled for a rich set of multidimensional covariates, there may still be endogeneity issues caused by reverse causality and omitted variable bias, such as family inherent risk preferences. As a sensitivity analysis, this study uses "Community Mean Expectation Gap" as an instrumental variable (IV) to capture the association between local peer pressure and individual educational expectations.

The theoretical basis of the instrumental variable is the local spatial characteristics of educational competition, which is often driven by Neighbourhood Effects and school district boundaries. Parents living in communities with high average educational expectations may be exposed to a strong atmosphere of intensive parenting and academic competition. This local peer pressure may increase the educational expectations of individual parents for their children, thereby supporting the relevance condition of the instrumental variable. In addition, the community-level aggregate expectation does not directly determine the internal reproductive decision within a specific family. It mainly plays a role through the channel of induced educational anxiety, which is intended to support the exclusion restriction of the instrumental variable.

The Two-Stage Least Squares (2SLS) estimates support the relevance of the instrumental variable and are broadly consistent with the baseline association. In the first stage of regression, instrumental variables show strong explanatory power for endogenous explanatory variables. The first stage F statistic is as high as 267.2, far exceeding the rule-of-thumb threshold of 10, which suggests that weak-instrument concerns are limited. The regression results of the second stage show that the core explanatory variable coefficient adjusted by IV is still highly significant at the 1% level, and its absolute value rises sharply from -0.063 of the benchmark OLS to -0.183. This result is important. It provides suggestive evidence that the education-expectation gap is associated with a decline in fertility intention and suggests that the baseline OLS estimate may be conservative. However, the IV result should be interpreted cautiously due to the difficulty of fully verifying the exclusion restriction. Also, the exclusion restriction cannot be fully verified because community-level expectations may also be correlated with local school quality, housing prices, or neighborhood socioeconomic conditions.

5.5 Exploratory Analysis: Reverse Causality and Dynamic Feedback

Table 8: Logit Model for Second-Child Decision

Variable	Logit
Education-expectation gap	0.146*** (0.031)
Controls	
age	-0.027*** (0.003)
gender	0.106 (0.057)
Health status (1-5 scale)	-0.063** (0.021)
Job satisfaction (1-5 scale)	-0.032*** (0.009)
Hukou controls	Yes
Observations	5,859
AIC	7483.3
Null deviance	7714.3
Residual deviance	7463.3

*Notes: Logit coefficients reported with standard errors in parentheses. Dependent variable is second_child. Hukou controls are included but not reported. Marginal effects and t-test results can be added once available. * p < 0.10, ** p < 0.05, *** p < 0.01.*

Although the main analysis focuses on how educational expectations are associated with lower desire for additional children, the Logit model (Table 8) predicts the actual state of second childbearing and shows a significantly positive coefficient ($\beta = 0.146$, $p < 0.01$). This finding raises the possibility of reverse causality. It suggests that families that already have a second child may face subsequent dilution of family resources and intensification of sibling competition. One possible interpretation is that this reality could increase parents' anxiety about downward mobility, thus raising their educational expectations for children. This reverse possibility highlights the limitation of cross-sectional data in clarifying the time order between fertility behaviour and educational anxiety. It should therefore be treated as exploratory evidence that requires further investigation using longitudinal panel data.

6. Discussion and Implications

6.1 Summary of Core Findings

This study results show that the intergenerational gap in educational expectations among Chinese families is closely related to the gap in reproductive intentions. When parents' expectations for their children's educational achievements are significantly higher than their own, the gap in their willingness to have children is often smaller. This discovery indicates that the willingness to have more children is not only influenced by the direct cost of raising children, but also by the subjective pressure that parents bear for the success of their children's education.

Not only that, the results also show that this influence has obvious group differences. The negative impact of the intergenerational education-expectation gap is more prominent in families with low educational attainment and those with middle-income. Meanwhile, income heterogeneity does not follow a simple linear resource constraint model. On the contrary, it points to the squeeze on the middle class.

Only a relatively high level of the education-expectation gap significantly suppresses fertility intention. The mechanism analysis further suggests that educational expectation pressure may be transformed into fertility pressure through wealth crowding-out and time-energy occupation. Exploratory analysis suggests that there may be a dynamic relationship between fertility behavior and educational anxiety.

6.2 Beyond Absolute Costs: Subjective Mobility Anxiety and Fertility Suppression

The negative effect of the intergenerational education-expectation gap first reveals a decision-making logic at the family level. When parents consider whether to continue childbearing, they do not only calculate the direct costs of having an additional child, but also assess whether the family still has the ability to maintain existing educational commitments. The resources here include not only money, but also time, attention, educational planning and continuous companionship. When parents have already set higher educational goals for their children, continuing to have children may mean that these resources need to be redistributed among more children. Therefore, having an additional child is not merely "one more child" , but may also be understood as a pressure on existing educational commitments.

This pressure may be further magnified in the context of educational involution in China. Educational achievement is not only related to personal development, but is also often understood by families as an important path for upward mobility and avoiding downward mobility. When parents believe that their children must achieve a better status through education, the decision to have children is no longer merely a choice on family size, but is linked to the future mobility opportunities for the family. The greater the intergenerational education-expectation gap, the more parents worry that they cannot ensure the educational competitiveness of multiple children at the same time. This concern may further reduce the willingness to have children.

This discovery also expands the explanation on the quantity-quality trade-off. "Quality" usually refers to educational expenditure or other observable inputs in traditional discussions. However, the results of this study indicate that quality also includes parents' subjective standards for the educational status their children should achieve. What influences the willingness to have children is not only the actual resources invested by the family, but also the extent to which parents believe they must invest resources. Therefore, fertility suppression can be understood as a family's response to mobility competition: when parents believe it is difficult to ensure that more children simultaneously achieve the expected educational outcomes, reducing their willingness to have children becomes a risk-avoidance approach.

6.3 Unequal Reproductive Pressure: Middle-Class Squeeze and Cultural Capital Barriers

Surprisingly, income heterogeneity deviates from the strict linear resource constraints assumed in H2. On the contrary, it reveals the squeeze of the middle class, that is, the inhibitory effect is obvious in middle-income families, not just strongest in the lowest-income group. This might reflect an anxiety centered on class maintenance. For the middle-income families, children's education is not only related to the upward mobility of family but also to whether their current social status can be maintained. They often find it difficult to withdraw from educational competition, as withdrawal might mean that their children will be at a disadvantage in future competition. But they may not be able to absorb the costs of tutoring, school selection, extracurricular activities and intensive companionship as easily as high-income families. By contrast, low-income families may largely withdraw from elite education competition, so the expectation gap may not translate into the same level of statistically significant fertility suppression. Therefore, the core of the middle-class squeeze is not merely high costs, but rather that households have to avoid downward mobility while being constrained by relatively rigid resource boundaries.

The stronger effect among lower-educated parents may reflect another kind of pressure. For such families, a larger intergenerational education-expectation gap means that children need to cross a longer educational distance. The higher the educational expectations, the higher the marginal mobility costs that families need to bear. However, due to the relatively limited available cultural capital, educational information and educational resources, low-education parents may need to make more compensatory efforts when supporting their children in achieving an educational leap. Therefore, when the expectation gap increases, the pressure of educational competition is more likely to be perceived as an additional realistic burden and further compress the space for continued childbearing.

These results jointly indicate that educational involution does not create exactly the same kind of family anxiety. Middle-income families are more likely to face the pressure of maintaining social class, while low-education families are more likely to face the pressure of upward mobility. The income finding also suggests that weaker resources do not automatically produce stronger fertility suppression, because some low-income families may have already stepped back from the most intense forms of educational competition. The social positions of these groups are different, but they both point to the same outcome: when the tension between educational expectations and the boundary of family resources increases, continuing to have children is more likely to be regarded as a risk of weakening children's educational opportunities.

However, this interpretation should remain bounded. There are differences within both middle-income families and low-education families. Different families may have different educational aspirations, resource conditions and fertility strategies. The results of this study indicate that the fertility-suppressing effect of the intergenerational education-expectation gap shows significant class differences, rather than suggesting that all families will respond to educational competition in the same way.

6.4 Dynamic Feedback Loop: Fertility Behavior and Educational Anxiety

Exploratory results suggest that the relationship between fertility behavior and educational anxiety may not be one-directional. The main analysis of this study indicates that a higher intergenerational education-expectation gap is associated with lower additional fertility intention. However, the second-child model also suggests that existing fertility behavior may coexist with a higher education-expectation gap, which provides an exploratory clue for reverse effects or dynamic feedback. This means that educational anxiety may not occur only before childbearing decisions; it may also be readjusted after a family has already formed a certain number of children.

One possible explanation is that families with multiple children may reorganize the allocation of internal resources. Parents need to allocate money, time, attention and educational opportunities among their children, so comparisons among siblings may also increase parents' uncertainty about educational outcomes. In this case, parents may respond to the dispersion of resources by raising their educational expectations.

Therefore, this dynamic perspective indicates that the family is not a static decision-making unit. Fertility decisions and educational expectations may both change as children grow up, enter school, and face more specific educational competition. Therefore, fertility behavior and educational anxiety may co-evolve throughout the family life cycle. This explanation should be understood as a suggestive interpretation rather than a mechanism that has been fully confirmed.

This dynamic relationship also implies that different families may not respond to educational competition in the same way. Some families may concentrate resources to raise fewer children, while others may understand additional children as a way to increase future mobility possibilities. From this perspective, fertility decisions under educational competition may not follow a single quantity-quality logic, but this possibility requires further evidence beyond the current cross-sectional data.

6.5 Policy Implications: Reducing Relative Anxiety and Structural Constraints

The findings of this study first challenge the default understanding of low fertility in many pronatalist policies.

Current policies often view economic burdens as the cause of low fertility rates, so the policy focus is concentrated on tools such as cash subsidies, childcare subsidies and tax incentives. These measures are necessary, but the results of this study show that families are reluctant to continue having children not only because they "cannot afford another child", but also because they are worried about not being able to provide sufficient educational support for each child. Actions of fertility promotion policies that only address direct costs often underestimate the inhibitory effect of educational competition on fertility intentions.

Not only that, but the deeper issue also actually lies in the fact that the social standards for "qualified parents" are constantly rising. Educational competition no longer merely demands that parents meet their children's basic growth needs, but requires them to continuously invest in educational planning, learning management and competitive resources. Having a child thus not only increases the responsibility of daily care, but also means shouldering an increasingly high set of parenting standards. When such standards are internalized within families, even if some economic costs decline, parents may still believe that having new children will weaken their ability to maintain high-quality parenting.

This also provides insight into the limitations of educational burden-reduction policies such as the "Double Reduction" policy. The issue is not merely the specific form of off-campus training, but rather that competitive pressure itself has a tendency to reproduce itself. As long as educational achievements remain an important basis for families to judge future opportunities, parents will continue to seek ways to maintain a competitive edge. After explicit tutoring is reduced, competitive pressure may shift to other resources and actions that families can mobilize. Therefore, alleviating fertility anxiety cannot merely rely on reducing one form of educational investment. It is also necessary to understand why educational competition is constantly being reorganized by families.

Time and energy pressure should also be understood within this structure. Under educational competition, parental participation has increasingly been regarded as a component of "qualified parenting". Supervising homework, arranging study, paying attention to pathways to further education, and maintaining communication with schools are not only practical tasks but also social expectations. For working parents, this intensive parenting norm may continue to conflict with work responsibilities. Therefore, the burden of time and energy is not merely a matter of “insufficient time”, but rather the pressure on parents to be both workers and intensive educational managers at the same time.

Therefore, the policy implications of this study do not lie in denying economic support, but in pointing out the interpretive boundary of low-fertility governance. When educational achievements are continuously regarded as an important path to social security and future opportunities, educational competition will be internalized by families as long-term reproductive pressure. In other words, the issue of low fertility is not only about the cost of raising children, but also about families’ persistent concerns about educational success and future security.

6.6 Limitations and Future Research

This study still has several limitations. First, this study uses cross-sectional data. Although the model controlled for a series of individual, family and regional variables and conducted robustness tests, cross-sectional data make it difficult to fully identify the temporal sequence between educational expectations and fertility intentions. Therefore, this study is more suitable for explaining the association between the two and its social implications, rather than being understood as definitive proof of a complete causal process.

Second, instrumental variable analysis can help alleviate some endogeneity problems, but it still has limitations. The average education-expectation gap at the community level can reflect the local educational atmosphere, but the exclusion restriction is difficult to fully verify. Community-level educational expectations may also be related to school quality, housing prices, neighborhood norms and local socio-economic conditions, and these factors may also influence fertility intentions through other pathways. Therefore, the IV results are more suitable as suggestive evidence rather than definitive causal evidence.

Third, the measurement of subjective educational expectations itself also has limitations. Parents’ expectations for their children’s educational achievements may simultaneously encompass educational aspirations, realistic plans and anxiety, and may also be influenced by the current number of children and their academic performance. The variables used in this study can capture an important dimension of subjective educational stress, but they still cannot fully distinguish these different meanings. Future research can use qualitative interviews to further understand whether parents express hope, stress, realistic judgment or class anxiety when reporting high educational expectations.

Overall, future research can be advanced in several directions. Firstly, longitudinal data can help researchers track changes in educational expectations and reproductive intentions within the same family. Secondly, the fixed-effect model can better control the unobserved characteristics at the household level. Thirdly, qualitative research can further reveal how parents understand educational success, mobility anxiety and fertility decisions in daily life. Finally, future research can further examine whether different families have adopted different mobility strategies, such as concentrating resources on raising several children or having more children, in order to disperse the uncertainties of future mobility. In other words, it is the idea that “having more children will surely lead to one success.” These directions may help to explain more comprehensively the dynamic relationship between educational competition and family population decisions.

7. Conclusion

This study explores how the intergenerational gap in educational expectations affects Chinese parents' willingness to have additional children when they already have at least one. Against the backdrop of low fertility rates and educational competition, this study shifts the focus of its analysis from the objective cost of raising children to a more relative pressure, that is, the distance between parents' own educational status and their educational expectations for their children. This view reveals an important issue: families' hesitation in having children not only stems from the current economic burden but also from their judgment of their children's future competitive position.

The empirical results of this study show that the larger the intergenerational education-expectation gap is, the lower additional fertility intention is. In other words, the reason why educational pressure affects fertility is not merely because educational investment is expensive, but because education has been understood by families as an important path for intergenerational mobility. Having another child is no longer merely a matter of adding another child; it also means taking on a long-term responsibility for future competition.

The heterogeneity results further indicate that this kind of pressure has a distinct social distribution. The income results do not follow a simple linear resource-constraint pattern, but point to a middle-class squeeze; meanwhile, parents with low educational attainment are also more likely to be affected by the gap in educational expectations. This indicates that educational pressure does not act on all existing-parent families in the same way. Educational competition is transformed into unequal reproductive pressure through different resource conditions, class positions, and mobility expectations. Mechanism analysis and exploratory results also point in the same direction: educational competition has gradually been embedded in families' daily arrangements and reproductive decision-making about additional children, and there may also be a dynamic relationship between additional childbearing behavior and educational anxiety that changes with the family life cycle.

Overall, this study reinterprets educational pressure from a simple cost issue as a kind of pressure related to relative position and mobility risks. By constructing the intergenerational education-expectation gap, this study attempts to transform subjective educational pressure that is difficult to observe directly into a testable empirical variable. This analysis also indicates that understanding China’s low fertility cannot merely rely on whether existing-parent families have sufficient resources; it is also necessary to see how families assess whether these resources can support their children in maintaining their positions in competition.

This study ultimately explains that China’s low fertility problem is not only an economic burden issue but also a mobility insecurity issue. When educational achievement is still regarded as an important path to obtaining stable opportunities and social positions, having another child may be understood by families as a long-term and uncertain competitive commitment. What restrains additional fertility intention is not only “how much it costs to raise a child” but also parents’ judgment of whether they can help their children not fall behind in competition. Low additional fertility intention among existing parents thus reflects families’ more cautious choices when facing future risks, class positions, and intergenerational responsibilities.

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THE UNIVERSITY OF HONG KONG
FACULTY OF SOCIAL SCIENCES
MASTER OF SOCIAL SCIENCES IN THE FIELD OF POPULATION AND POLICY ANALYSIS

Assignment Submission Declaration Form

Group Number:	Group 8
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Name of Supervisor:	Guy Abel
Title of Assignment:	The Price of "Success of Children": How Does Intergenerational Education-Expectation Gap Affect Fertility Intention
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